



Tobias Rosenbaum

Sr. Manager Computer System Validation & Data Integrity

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Profile

Senior IT Lead with 13+ years in Computerized System Validation and Data Integrity in GMP-regulated pharma. Key role in transforming BioNTech from Startup to Enterprise by enhancing CSV/DI governance and audit readiness. Achieved global QC-LIMS rollout in 7 months across 3 sites, supporting €19bn revenue. Effective leader with fluent DE/EN skills, collaborates with QA, Manufacturing, and vendors to streamline validation activities.

Professional Experience

Dec 2024 – present
Mainz, Germany

ASSOCIATE DIRECTOR E2E SUPPLY CHAIN SYSTEMS, BIONTECH SE, *BIONTECH SE*

Lead cross-functional teams delivering cutting-edge IT systems for individualized cancer treatments, including Chain of Identity (COI) and Chain of Custody (COC) compliance.

- Lead a team of system owners responsible for the complete system lifecycle including validation activities and cross-functional partnerships.
- Manage an AI automation project using Databricks and large language models to streamline regulatory document generation, aiming for over 60% efficiency improvement while establishing AI governance frameworks for compliance-sensitive environments.
- Act as Domain Architect for the End-to-End Supply Chain Management Area, defining target architecture and necessary transformations, including budget estimation.
- Oversee governance processes ensuring alignment between business and technical teams.
- Guided team development, recruitment, and organizational goal alignment.

May 2021 – Nov 2024
Mainz, Germany

ASSOCIATE DIRECTOR DIGITAL SOLUTIONS QUALITY SYSTEMS ARCHITECTURE, *BIONTECH SE*

Led team of architects developing enterprise-scale IT solutions for quality control, quality assurance and clinical labs.

- Organised cross department IT support for facility engineering runs after major system updates, leading to efficient handling of issues and a completion of engineering runs in time.
- Defined cross-domain process and data flow models enabling traceability and ALCOA+ principles end to end.
- Implemented ISO 25010-based quality framework for vendor selection and system design.
- Took over leading role in the company's FDA audit preparation program's system workstream, leading to zero critical system related findings in the following audits.

Oct 2018 – May 2021
Mainz, Germany

SYSTEM ENGINEER, BIONTECH SE

Established processes for GxP compliant applications and led major implementation projects.

- Led record-breaking global QC LIMS implementation (7 months, 3 sites) as solution architect, directly enabling quality control processes supporting €19 billion revenue impact, incl. validation strategy, DI controls (e.g., audit trail configuration, access management, backup and recovery)
- Established GxP-compliant application processes, advancing to acting team lead.
- Led GCLP application team through critical transition period during COVID vaccine development
- Standardized change control and release validation across LIMS, increasing consistency of reviews and DI documentation.

Aug 2012 – Sep 2018
Wiesentheid, Germany

PROFESSIONAL SYSTEM ENGINEER,

Bayerischer Blutspendedienst gGmbH

and Managed full lifecycle of 30+ applications processing 7,500 blood products daily, pioneering data-driven insights.

- Integrated Manufacturing Execution Systems with manufacturing and laboratory equipment maintaining GMP standards.
- Served as IT Quality Officer, representing the company in both internal and external system related audits.
- Managed full lifecycle of software projects for over 30 key applications that process 7,500 blood products daily, including investigating deviations and owning system related CAPAs.
- Executed QC-LIMS implementation with detailed V-model validation, setting a new vendor benchmark.

Jun 2011 – Jul 2012
Penzberg, Germany

SYSTEM ANALYST, ROCHE DIAGNOSTICS GMBH

- Validated MES for Herceptin production across 4 facilities (test design/execution, documentation review).
- Created unified, validated dashboard aggregating data from 4 MES systems
- System Owner for OSIsoft PI and Roche Data Display (batch release decision support), ensuring GMP compliance and data integrity across manufacturing data flows.

Education

Oct 2005 – Feb 2011
Würzburg, Germany

German Diploma (Biology/Life Sciences),

Julius-Maximilians-Universität

Focus: Computational Biology, Neurobiology, Bioinformatics Thesis: Pattern recognition in brain waves of honey bees with Matlab

Sep 2002 – Jan 2005
Frankfurt a.M., Germany

Computer System Developer, Provadis Partner für Bildung GmbH

Completed 3-year program in 2.5 years due to exceptional performance

1992 – 2001
Mainz, Germany

Allgemeine Hochschulreife (Abitur), Theresianum Gymnasium Mainz

Examination subjects: biology, social studies, german, mathematics

Projects

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| 2022 – 2023 | Clinical Lab Blueprint, Domain Architect <ul style="list-style-type: none">• Redesigned software landscape for individualized mRNA vaccine production• Reduced system landscape from 25 legacy systems to 4 central domain-specific systems• Defined integration standards improving data integrity and reducing turnaround time |
| 2020 – 2021 | BioNTech Global QC LIMS Implementation, Solution Architect <ul style="list-style-type: none">• Coordinated validation strategy across manufacturer, QA, validation manager, and IT teams• Delivered project in 7 months from start to multi-site rollout• Enabled quality control for increasing Comirnaty production, supporting €19bn revenue• Established company standards for validation documentation |
| 2014 – 2017 | QC LIMS for Bavarian Blood Donation Service, Project Lead <ul style="list-style-type: none">• Replaced legacy application with modern QC-LIMS for 6,000 daily batch analyses• Completed project within time and budget while streamlining processes• Coordinated automated analyzer connections and test plan execution |

Core Competencies

GxP Compliance & Computer System Validation (13+ years)	● ● ● ● ●	Team Leadership & Cross Functional Project Management	● ● ● ● ●
Stakeholder Management & C-Level Consulting	● ● ● ● ●	Laboratory Informatics & LIMS Implementation	● ● ● ● ●
Regulatory Audit Preparation & Quality Assurance	● ● ● ● ●	Digital Strategy & System Architecture	● ● ● ● ●
Software and Database Development	● ● ● ● ●		

Certificates

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| • Expert for Computersystemvalidation | • Certified Professional Software Architect Foundation Level |
| • ITIL Foundation Level | |

Languages

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| • German | • English |
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Interests

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| • Photography | • Woodworking and DIY | • Agentic Systems |
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References

Jürgen Richert, Vice President IT, Immatics

Christoph Eller, Director Digital Solutions Quality, BioNTech SE